Notes on Gauge Theory

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Contents

2 Associated Bundles and Covariant Derivative

2.1 Associated Bundles

Let $\pi : P \to M$ be a principal G-bundle, F be a smooth manifold with a left G-action $\rho : G \to \mathrm{Diff}(F)$.

Definition 2.1. The **associated bundle** of P with fiber F is the quotient space

$$E := P \times_\rho F := (P \times F)/\sim,$$

where the equivalence relation $\sim$ is defined by

$$(p, f) \sim (pg, g^{-1}f), \quad \forall p \in P, f \in F, g \in G.$$

Denote the equivalence class of (p, f) by $[p, f]$. We can define a projection map $\pi_E : E \to M$ by $\pi_E([p, f]) = \pi(p)$. Since

$$\pi_E([pg, g^{-1}f]) = \pi(pg) = \pi(p) = \pi_E([p, f]),$$

π_E is well-defined.

Fact 2.1. The associated bundle $E := P \times_\rho F$ with the projection π_E gives a fiber bundle over M with fiber F. Let $\{(U_\alpha, \phi_\alpha)\}$ be a local trivialization of P

$$\phi_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times G, \quad p \mapsto \Big(\pi(p), g_\alpha(p)\Big),$$

The definition of $g_\alpha(p)$ is as follows: Let $s_\alpha : U_\alpha \to P$ be a local section defined by $s_\alpha(x) = \phi_\alpha^{-1}(x, e)$. Since the action of G on P is free and transitive on each fiber, for any $p \in \pi^{-1}(U_\alpha)$, there exists a unique $g_\alpha(p) \in G$ such that $p = s_\alpha(\pi(p))g_\alpha(p)$. The transition functions of this local trivialization are given by

$$g_{\alpha\beta}(x) = g_\alpha(s_\beta(x)).$$

Then we can construct a local trivialization $\{(U_\alpha, \psi_\alpha)\}$ of E by

$$\psi_\alpha : \pi_E^{-1}(U_\alpha) \to U_\alpha \times F, \quad [p, f] \mapsto \Big(\pi(p), \rho(g_\alpha(p))f\Big).$$

One can check that ψ_α is well-defined and is a diffeomorphism. The transition functions of this local trivialization are

$$\rho \circ g_{\alpha\beta} : U_\alpha \cap U_\beta \to \mathrm{Diff}(F).$$

Example 2.2. If $P = M \times G$ is a trivial, then $P \times_\rho F$ is trivial. If the action ρ is trivial, then $P \times_\rho F$ is also trivial.

Example 2.3. If $F = V$ is a finite-dimensional vector space and $\rho : G \to \mathrm{GL}(V)$ is a linear representation of G, then the associated bundle $E = P \times_\rho V$ is a vector bundle over M with fiber V. The vector bundle structure is given by

$$[p, v] + [p, w] := [p, v + w], \quad \lambda[p, v] := [p, \lambda v], \quad \forall p \in P, v, w \in V, \lambda \in \mathbb{R}.$$

One can check that the vector bundle structure is well-defined.

Example 2.4. Let $P = \mathrm{Fr}(E)$ be the frame bundle of a rank m vector bundle E over M.

- $V = \mathbb{R}^m$, $\mathrm{GL}(m, \mathbb{R}) \curvearrowright \mathbb{R}^m : g \cdot v = gv$, we get $\mathrm{Fr}(E) \times_{\rho_{\mathrm{can}}} \mathbb{R}^m \cong E$.
- $V = \mathbb{R}^{m\times m}$, $\mathrm{GL}(m, \mathbb{R}) \curvearrowright \mathbb{R}^{m\times m} : g \cdot A = gAg^{-1}$, we get $\mathrm{End}(E)$.
- $V = \bigwedge^k(\mathbb{R}^m)^*$, $\mathrm{GL}(m, \mathbb{R}) \curvearrowright \bigwedge^k(\mathbb{R}^m)^* : g \cdot \alpha(v_1, \dots, v_k) = \alpha(g^{-1}v_1, \dots, g^{-1}v_k)$, we get $\bigwedge^k E^*$.
- $V = \mathrm{Sym}^k(\mathbb{R}^m)^*$, $\mathrm{GL}(m, \mathbb{R}) \curvearrowright \mathrm{Sym}^k(\mathbb{R}^m)^* : g \cdot \alpha(v_1, \dots, v_k) = \alpha(g^{-1}v_1, \dots, g^{-1}v_k)$, we get $\mathrm{Sym}^k E^*$.
- $V = \mathbb{R}$, $\mathrm{GL}(m, \mathbb{R}) \curvearrowright \mathbb{R} : g \cdot r = (\det g)r$, we get $\det E$.

Example 2.5. Take $F = G$ and $\rho : G \to \mathrm{Aut}(G) : \rho(g)(h) = ghg^{-1}$ the conjugation action, then the associated bundle $P \times_{\mathrm{conj}} G$ is denoted by $\mathrm{Ad}P$. Take $V = \mathfrak{g}$ and $\rho = \mathrm{Ad} : G \to \mathrm{GL}(\mathfrak{g})$ the adjoint representation. The associated bundle $\mathrm{ad}P := P \times_{\mathrm{Ad}} \mathfrak{g}$ is called the **adjoint bundle** of P.

Over this section, we mainly focus on the case where $F = V$ and $\rho : G \to \mathrm{GL}(V)$ is a linear representation of G.

Fix $x \in M$, let P_x be the fiber of P over x and E_x be the fiber of E over x. Then for every $p \in P_x$, there is a canonical way to identify E_x with V by

$$f_p : V \to E_x, \quad f_p(v) = [p, v].$$

Lemma 2.6. *The map f_p is a linear isomorphism.*

Proof. Injectivity: If $[p, v] = [p, w]$, then there exists $g \in G$ such that $pg = p$ and $g^{-1}v = w$. Since the action of G on P is free, we have $g = e$ and thus $v = w$.

Surjectivity: For any $[q, w] \in E_x$, $q \in P_x$, so there exists $g \in G$ such that $q = pg$. Then we have $[q, w] = [pg, w] = [p, gw] = f_p(gw)$. □

Lemma 2.7. *Let $f_p : V \to E_x$ be the map defined above. For any $g \in G$, $f_{pg} = f_p \circ \rho(g)$.*

Proof. For any $v \in V$, we have

$$f_{pg}(v) = [pg, v] = [p, gv] = f_p(gv) = f_p \circ \rho(g)(v).$$ □

2.2 Tensorial Forms on a Principal Bundle

Let $\pi : P \to M$ be a principal G-bundle, $\rho : G \to \mathrm{GL}(V)$ be a finite-dimensional representation of G, and $E = P \times_\rho V$ be the associated vector bundle.

Definition 2.8. A V-valued k-form $\varphi \in \Omega^k(P, V)$ is called **tensorial** if it satisfies the following two conditions:

- φ is **horizontal**: $\varphi(X_1, \cdots, X_k) = 0$ if at least one of X_i is vertical.
- φ is **G-equivariant of type** ρ: $r_g^*\varphi = \rho(g^{-1}) \cdot \varphi$ for all $g \in G$.

The set of all tensorial k-forms on P of type ρ is denoted by $\Omega^k_\rho(P, V)$.

Example 2.9. The curvature Ω of a connection ω on P is a tensorial 2-form of type Ad.

Below we will show that there is a one-to-one correspondence between $\Omega^k_\rho(P,V)$ and $\Omega^k(M,E)$, where $\Omega^k(M,E)$ is the space of E-valued k-forms on M.

Definition 2.10. Given $x \in M$ and $X_1,\cdots,X_k \in T_xM$, choose any point $p \in P_x$ and choose any lift $\tilde{X}_1,\cdots,\tilde{X}_k \in T_pP$ of $X_1,\cdots,X_k$, i.e., $\tilde{X}_i \in T_pP$ and $\pi_*(\tilde{X}_i) = X_i$. Define $\varphi^\flat$ by

$$\varphi^\flat_x(X_1,\cdots,X_k) := f_p(\varphi_p(\tilde{X}_1,\cdots,\tilde{X}_k)) \in E_x.$$

where $f_p : V \to E_x$, $v \mapsto [p,v]$ is the canonical isomorphism defined above.

Conversely, given $\psi \in \Omega^k(M,E)$, $p \in P$ and $Y_1,\cdots,Y_k \in T_pP$, define $\psi^\sharp$ by

$$\psi^\sharp_p(Y_1,\cdots,Y_k) := f_p^{-1}\big(\psi_{\pi(p)}(\pi_*Y_1,\cdots,\pi_*Y_k)\big) \in V.$$

Theorem 2.11. *These two maps*

$$\begin{aligned} \Omega^k_\rho(P,V) &\to \Omega^k(M,E), \quad \varphi \mapsto \varphi^\flat, \\ \Omega^k(M,E) &\to \Omega^k_\rho(P,V), \quad \psi \mapsto \psi^\sharp, \end{aligned}$$

are well-defined and inverse to each other, thus we get linear isomorphisms between $\Omega^k_\rho(P,V)$ *and* $\Omega^k(M,E)$.

Before proving the theorem, let's see some examples and corollaries.

Example 2.12. By the theorem above, the curvature Ω of a connection on a principal G-bundle P can be viewed as an element in $\Omega^2(M,\mathrm{ad}P)$, a 2-form on M with values in the adjoint bundle $\mathrm{ad}P$.

Corollary 2.13. *Take $k=0$ in Theorem 2.11, an element in $\Omega^0_\rho(P,V)$ is just a G-equivariant smooth map $f : P \to V$ of type ρ, i.e.,*

$$f(pg) = (r_g^*f)(p) = \rho(g^{-1})f(p) = g^{-1}\cdot f(p), \quad \forall p \in P, g \in G.$$

Besides, $\Omega^0(M,E)$ is just the space of smooth sections of E. Hence we get a one-to-one correspondence

$$\left\{ \begin{array}{c} G\text{-equivariant maps w.r.t. } \rho \\ f : P \to V \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{c} \text{sections of the associated bundle} \\ s : M \to P\times_\rho V \end{array} \right\}$$

Proof of Theorem 2.11. We first show that $\varphi^\flat$ is well-defined. First check that $\varphi^\flat$ does not depend on the choice of the lift $\tilde{X}_i$. If $\tilde{X}'_i$ is another lift of X_i, then $\pi_*(\tilde{X}_i - \tilde{X}'_i) = 0$, so $\tilde{X}_i - \tilde{X}'_i$ is vertical. Since φ is horizontal,

$$\varphi_p(\tilde{X}'_1,\cdots,\tilde{X}'_k) = \varphi_p(\tilde{X}_1 + \text{vertical},\cdots,\tilde{X}_k + \text{vertical}) = \varphi_p(\tilde{X}_1,\cdots,\tilde{X}_k).$$

Next check that $\varphi^\flat$ does not depend on the choice of $p \in P_x$. Suppose $p' = pg$ for some $g \in G$, then $r_{g*}\tilde{X}_i$ is a lift of X_i at pg. Since φ is G-equivariant of type ρ, we have

$$\varphi_{pg}(r_{g*}\tilde{X}_1,\cdots,r_{g*}\tilde{X}_k) = (r_g^*\varphi_{pg})(\tilde{X}_1,\cdots,\tilde{X}_k) = \rho(g^{-1})\cdot\varphi_p(\tilde{X}_1,\cdots,\tilde{X}_k).$$

By Lemma 2.7,

$$f_{pg}(\varphi_{pg}(r_{g*}\tilde{X}_1, \cdots, r_{g*}\tilde{X}_k)) = f_p \circ \rho(g)(\rho(g^{-1}) \cdot \varphi_p(\tilde{X}_1, \cdots, \tilde{X}_k)) = f_p(\varphi_p(\tilde{X}_1, \cdots, \tilde{X}_k)).$$

Thus $\varphi^\flat$ is well-defined.

Let $\psi \in \Omega^k(M, E)$, it's easy to see that $\psi^\sharp$ is horizontal. For G-equivariance, we have

$$\begin{aligned}
(r_g^*\psi^\sharp)_p(Y_1, \cdots, Y_k) &= \psi^\sharp_{pg}(r_{g*}(Y_1), \cdots, r_{g*}(Y_k)) \\
&= f_{pg}^{-1}\big(\psi_{\pi(pg)}(\pi_* r_{g*}(Y_1), \cdots, \pi_* r_{g*}(Y_k))\big) \\
&= f_{pg}^{-1}\big(\psi_{\pi(p)}(\pi_* Y_1, \cdots, \pi_* Y_k)\big) \\
&= \rho(g^{-1}) \cdot f_p^{-1}\big(\psi_{\pi(p)}(\pi_* Y_1, \cdots, \pi_* Y_k)\big) \\
&= \rho(g^{-1}) \cdot \psi^\sharp_p(Y_1, \cdots, Y_k).
\end{aligned}$$

Hence $\psi^\sharp$ is G-equivariant of type ρ, and thus $\psi^\sharp \in \Omega^k_\rho(P, V)$.

Finally, we check that these two maps are inverse to each other. For any $\varphi \in \Omega^k_\rho(P, V)$, $p \in P$ and $Y_1, \cdots, Y_k \in T_pP$,

$$\begin{aligned}
(\varphi^{\flat\sharp})_p(Y_1, \cdots, Y_k) &= f_p^{-1}\big(\varphi^\flat_{\pi(p)}(\pi_* Y_1, \cdots, \pi_* Y_k)\big) \\
&= f_p^{-1}\big(f_p(\varphi_p(Y_1, \cdots, Y_k))\big) \\
&= \varphi_p(Y_1, \cdots, Y_k).
\end{aligned}$$

For any $\psi \in \Omega^k(M, E)$, $x \in M$ and $X_1, \cdots, X_k \in T_xM$, choose any $p \in P_x$ and any lift $\tilde{X}_1, \cdots, \tilde{X}_k \in T_pP$ of $X_1, \cdots, X_k$,

$$\begin{aligned}
(\psi^{\sharp\flat})_x(X_1, \cdots, X_k) &= f_p(\psi^\sharp_p(\tilde{X}_1, \cdots, \tilde{X}_k)) \\
&= f_p\big(f_p^{-1}(\psi_x(\pi_*\tilde{X}_1, \cdots, \pi_*\tilde{X}_k))\big) \\
&= \psi_x(X_1, \cdots, X_k).
\end{aligned}$$

□

2.3 Covariant Derivative

Keep the same notations as above. Notice that given a horizontal distribution H on P, we can decompose the tangent vetor $Y_p \in T_pP$ into a horizontal part and a vertical part,

$$Y_p = hY_p + vY_p, \quad hY_p \in H_p, vY_p \in V_pP.$$

In this section, we will use this decomposition to define the covariant derivative of vector-valued forms on P, and thus a covariant derivative on the associated bundle E.

Let's start with the usual exterior derivative $\mathrm{d} : \Omega^k(P, V) \to \Omega^{k+1}(P, V)$.

Proposition 2.14. *If $\varphi \in \Omega^k(P, V)$ is G-equivariant of type ρ, then $\mathrm{d}\varphi$ is also G-equivariant of type ρ.*

Proof. For any $g \in G$,

$$r_g^*(\mathrm{d}\varphi) = \mathrm{d}(r_g^*\varphi) = \mathrm{d}(\rho(g^{-1})\varphi) = \rho(g^{-1})\mathrm{d}\varphi.$$

The last equality holds since $\rho(g^{-1})$ is a constant linear map on V for a fixed g. □

In general, the exterior derivative does not preserve the horizontality, so it does not preserve the space of tensorial forms.

Definition 2.15. For any $\varphi \in \Omega^k(P, V)$, we define its **horizontal component** $\varphi^h \in \Omega^k(P, V)$ by

$$\varphi^h_p(Y_1, \cdots, Y_k) := \varphi_p(hY_1, \cdots, hY_k),$$

for any $p \in P$ and $Y_1, \cdots, Y_k \in T_pP$.

Proposition 2.16. *If $\varphi \in \Omega^k(P, V)$ is G-equivariant of type ρ, so is φ^h.*

Proof. For $g \in G$, $p \in P$ and $Y_1, \cdots, Y_k \in T_pP$,

$$\begin{aligned}
(r_g^*\varphi^h)_p(Y_1, \cdots, Y_k) &= \varphi^h_{pg}(r_{g*}Y_1, \cdots, r_{g*}Y_k) \\
&= \varphi_{pg}(h(r_{g*}Y_1), \cdots, h(r_{g*}Y_k)) \\
&= \varphi_{pg}(r_{g*}(hY_1), \cdots, r_{g*}(hY_k)) \\
&= (r_g^*\varphi)_p(hY_1, \cdots, hY_k) \\
&= \rho(g^{-1}) \cdot \varphi_p(hY_1, \cdots, hY_k) \\
&= \rho(g^{-1}) \cdot \varphi^h_p(Y_1, \cdots, Y_k).
\end{aligned}$$

□

Proposition 2.14 and 2.16 together imply that if $\varphi \in \Omega^k(P, V)$ is G-equivariant of type ρ, then $(\mathrm{d}\varphi)^h$ is a tensorial form.

Definition 2.17. For any $\varphi \in \Omega^k(P, V)$, we define its **covariant derivative** by

$$D\varphi := (\mathrm{d}\varphi)^h.$$

In particular, D maps tensorial k-forms of type ρ to tensorial $(k+1)$-forms of type ρ, i.e.,

$$D : \Omega^k_\rho(P, V) \to \Omega^{k+1}_\rho(P, V).$$

Proposition 2.18. *The covariant derivative $D : \Omega^k_\rho(P, V) \to \Omega^{k+1}_\rho(P, V)$ is an antiderivation of degree $+1$, i.e., for any $\varphi \in \Omega^k_\rho(P, V)$ and $\psi \in \Omega^l_\rho(P, V)$,*

$$D(\varphi \wedge \psi) = D\varphi \wedge \psi + (-1)^k \varphi \wedge D\psi.$$

Proof. Since φ and ψ are horizontal, we have

$$\begin{aligned}
D(\varphi \wedge \psi) = (\mathrm{d}(\varphi \wedge \psi))^h &= (\mathrm{d}\varphi \wedge \psi + (-1)^k \varphi \wedge \mathrm{d}\psi)^h \\
&= (\mathrm{d}\varphi)^h \wedge \psi^h + (-1)^k \varphi^h \wedge (\mathrm{d}\psi)^h \\
&= D\varphi \wedge \psi + (-1)^k \varphi \wedge D\psi.
\end{aligned}$$

□

Example 2.19. By Theorem 2.11, the curvature Ω of a connection on P is exactly the covariant derivative of ω, since

$$\Omega_p(X_p, Y_p) = \mathrm{d}\omega_p(hX_p, hY_p) = (\mathrm{d}\omega_p)^h(X_p, Y_p) = (D\omega_p)(X_p, Y_p).$$

By the one-to-one correspondence between $\Omega^k_\rho(P, V)$ and $\Omega^k(M, E)$, we can also define a covariant derivative on $\Omega^k(M, E)$

$$D : \Omega^k(M, E) \to \Omega^{k+1}(M, E), \quad D\psi := (D\psi^\sharp)^\flat.$$

Example 2.20. Let E be a rank m vector bundle over M, $\mathrm{Fr}(E)$ be its frame bundle. and ω be a connection on $\mathrm{Fr}(E)$ induced by a connection ∇ on E. With the identification $E = \mathrm{Fr}(E) \times_{\rho_{\mathrm{can}}} \mathbb{R}^m$, we claim that the covariant derivative D on $\Omega^k(M, E)$ is exactly ∇. The proof will be given later.

2.4 A Formula for the Covariant Derivative on Tensorial Forms

We have defined the covariant derivative D on $\Omega^k(P, V)$ by $D\varphi = (\mathrm{d}\varphi)^h$. In this section, we will give a more explicit formula for D, but only for $\Omega^k_\rho(P, V)$.

The representation $\rho : G \to \mathrm{GL}(V)$ induces a Lie algebra representation $\rho_* : \mathfrak{g} \to \mathrm{End}(V)$, which alows us to define a product of $\tau \in \Omega^k(P, \mathfrak{g})$ and $\varphi \in \Omega^l(P, V)$

Definition 2.21. For $\tau \in \Omega^k(P, \mathfrak{g})$ and $\varphi \in \Omega^l(P, V)$, we define $\tau \cdot \varphi \in \Omega^{k+l}(P, V)$ by

$$(\tau \cdot \varphi)_p(Y_1, \cdots, Y_{k+l}) := \frac{1}{k!l!} \sum_\sigma \mathrm{sgn}(\sigma) \cdot \rho_*\big(\tau_p(Y_{\sigma(1)}, \cdots, Y_{\sigma(k)})\big)\varphi_p(Y_{\sigma(k+1)}, \cdots, Y_{\sigma(k+l)}).$$

One can check that $\tau \cdot \varphi$ is indeed a $(k+l)$-form on P with values in V.

Example 2.22. If $V = \mathfrak{g}$ and $\rho = \mathrm{Ad}$, then

$$(\tau \cdot \varphi)_p(Y_1, \cdots, Y_{k+l}) = \frac{1}{k!l!} \sum_\sigma \mathrm{sgn}(\sigma) \cdot [\tau_p(Y_{\sigma(1)}, \cdots, Y_{\sigma(k)}), \varphi_p(Y_{\sigma(k+1)}, \cdots, Y_{\sigma(k+l)})].$$

In this case, we write $\tau \cdot \varphi$ instead of $[\tau, \varphi]$.

Theorem 2.23. *Let $\pi : P \to M$ be a principal G-bundle with a connection ω, $\rho : G \to \mathrm{GL}(V)$ be a finite-dimensional representation of G. If $\varphi \in \Omega^k_\rho(P, V)$, then*

$$D\varphi = \mathrm{d}\varphi + \omega \cdot \varphi.$$

Before proving the theorem, we frist see some examples.

Example 2.24. In case that $V = \mathfrak{g}$ and $\rho = \mathrm{Ad}$, for any $\varphi \in \Omega^k_{\mathrm{Ad}}(P, \mathfrak{g})$,

$$D\varphi = \mathrm{d}\varphi + [\omega, \varphi].$$

Notice that this formula only holds for tensorial forms. Since ω is not horizontal, we cannot use this formula to compute $D\omega$, in fact, we know that

$$D\omega = (\mathrm{d}\omega)^h = \Omega = \mathrm{d}\omega + \frac{1}{2}[\omega, \omega].$$

Since Ω is tensorial, we can apply the formula for Ω. In fact, we have

$$D\Omega = \mathrm{d}\Omega + [\omega, \Omega].$$

So the second Bianchi identity can be rewritten as $D\Omega = 0$.

Proof of Theorem 2.23. To verify the formula, we actually need to show

$$\mathrm{d}\varphi(hY_1, \cdots, hY_{k+1}) = \mathrm{d}\varphi(Y_1, \cdots, Y_{k+1}) + (\omega \cdot \varphi)_p(Y_1, \cdots, Y_{k+1}) =: \mathrm{I} + \mathrm{II}.$$

Since both sides are multilinear and antisymmetric in $Y_1, \cdots, Y_{k+1}$, we can decompose all the Y_i into horizontal and vertical parts and check the equality in several cases. Moreover, since we are verifying the equalityly, we can assume that the vertical part of $Y_i = \underline{A_i}$ is a fundamental vector field, and the horizontal part of $Y_i = \tilde{B}_i$ is a horizontal lift of some vector field B_i on M. Notice that $\tilde{B}_i$ is right-invariant.

Case 1: All Y_i are horizontal. Then $\mathrm{II} = 0$ since ω vanishes on horizontal vectors, and the formula holds trivially.

Case 2: At least two of the Y_i are vertical. Then

$$\begin{aligned}\mathrm{I} = \mathrm{d}\varphi(Y_1, \cdots, Y_{k+1}) = &\sum_i (-1)^{i+1} Y_i \varphi(Y_1, \cdots, \hat{Y}_i, \cdots, Y_{k+1}) \\ &+ \sum_{i<j} (-1)^{i+j} \varphi([Y_i, Y_j], Y_1, \cdots, \hat{Y}_i, \cdots, \hat{Y}_j, \cdots, Y_{k+1}).\end{aligned}$$

Since $[\underline{A_i}, \underline{A_j}] = \underline{[A_i, A_j]}$ is still vertical, at least one of the arguments is vertical, so $\mathrm{I} = 0$.

On the other hand,

$$\mathrm{II} = \frac{1}{k!} \sum_{\sigma \in S_{k+1}} \operatorname{sgn}(\sigma) \cdot \rho_*(\omega_p(Y_{\sigma(1)}))\varphi_p(Y_{\sigma(2)}, \cdots, Y_{\sigma(k+1)}),$$

which is also zero since at least one of the arguments of φ is vertical.

Case 3: Exactly one of the Y_i is vertical, say $Y_1 = \underline{A}$ and $Y_2, \cdots, Y_{k+1}$ are horizontal. Then

$$\mathrm{I} = \mathrm{d}\varphi(\underline{A}, Y_2, \cdots, Y_{k+1}) = \underline{A}\varphi(Y_2, \cdots, Y_{k+1}) + \sum_{i=2}^{k+1} (-1)^i \varphi([\underline{A}, Y_i], Y_2, \cdots, \hat{Y}_i, \cdots, Y_{k+1}).$$

In our assumption, Y_i is a horizontal lift of some vector field B_i on M, so is right-invariant. Hence $[\underline{A}, Y_i] = 0$ for $i = 2, \cdots, k+1$. Thus

$$\mathrm{I} = \underline{A}\varphi(Y_2, \cdots, Y_{k+1}).$$

On the other hand,

$$\begin{aligned}\mathrm{II} &= \frac{1}{k!} \sum_{\substack{\sigma \in S_{k+1} \\ \sigma(1)=1}} \operatorname{sgn}(\sigma) \cdot \rho_*(\omega(Y_{\sigma(1)}))\varphi(Y_{\sigma(2)}, \cdots, Y_{\sigma(k+1)}) \\ &= \rho_*(\omega(\underline{A}))\varphi(Y_2, \cdots, Y_{k+1}) = \rho_*(A)\varphi(Y_2, \cdots, Y_{k+1}).\end{aligned}$$

If we denote by f the function $\varphi(Y_2, \cdots, Y_{k+1})$. For $p \in P$, to calculate $\underline{A}_p f$, let $c(t) = \exp(tA)$ be the integral curve of A in G with $c(0) = e$. Then withe $j_p : G \to P, j_p(g) = pg$, we have

$$\underline{A}_p f = j_{p*}(A) f = j_{p*}(c'(0)) f = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} (f \circ j_p \circ c).$$

By the right-invariance of the horizontal vector fields $Y_2, \cdots, Y_{k+1}$,

$$\begin{aligned}(f \circ j_p \circ c)(t) &= f(pc(t)) = \varphi_{pc(t)}(Y_{2,pc(t)}, \cdots, Y_{k+1,pc(t)}) \\ &= \varphi_{pc(t)}(r_{c(t)*}Y_{2,p}, \cdots, r_{c(t)*}Y_{k+1,p}) \\ &= (r_{c(t)}^* \varphi)_p(Y_{2,p}, \cdots, Y_{k+1,p}) \\ &= \rho(c(t)^{-1})\varphi_p(Y_{2,p}, \cdots, Y_{k+1,p}) \\ &= \rho(\exp(-tA)) f(p).\end{aligned}$$

Thus

$$\underline{A}_p f(p) = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \rho(\exp(-tA)) f(p) = -\rho_*(A) f(p).$$

Hence $\mathrm{I} + \mathrm{II} = 0$. The left-hand side of the formula is also zero because $hY_1 = 0$. $\square$

Exercises

Exercise 2.1 (Forms on P pulled back from M)**.** Since $\pi : P \to M$ is a surjective submersion, the pull-back $\pi^* : \Omega^*(M,V) \to \Omega^*(P,V)$ is injective. A differential form $\varphi \in \Omega^*(P,V)$ is said to be **basic** if it is the pull-back $\pi^*\psi$ of some $\psi \in \Omega^*(M,V)$; it is said to be G**-invariant** if $r_g^*\varphi = \varphi$ for all $g \in G$. Show that:

- φ is G-invariant if and only if φ is G-equivariant of type ρ_{trivial}, where ρ_{trivial} is the trivial representation.
- φ is basic if and only if φ is horizontal and G-invariant.

Proof. If φ is G-equivariant of type ρ_{trivial}, then

$$r_g^*\varphi = \rho_{\text{trivial}}(g^{-1}) \cdot \varphi = \varphi, \quad \forall g \in G.$$

This is just the definition of G-invariance.

Since the associated bundle $E = P \times_{\rho_{\text{trivial}}} V$ is just the trivial bundle $M \times V$, and $f_p : V \to E_{\pi(p)} = V$ is the identity map, by Theorem 2.11,

$$\begin{aligned}\Omega^k(M,E) \cong \Omega^k(M, M\times V) = \Omega^k(M,V) &\xrightarrow{\cong} \Omega^k_{\rho_{\text{trivial}}}(P,V) = \{\text{basic forms on } P\}\\ \psi &\mapsto \psi^\sharp.\end{aligned}$$

where

$$\begin{aligned}\psi_p^\sharp(Y_1,\cdots,Y_k) &= f_p^{-1}\Big(\psi_{\pi(p)}(\pi_*Y_1,\cdots,\pi_*Y_k)\Big)\\ &= \psi_{\pi(p)}(\pi_*Y_1,\cdots,\pi_*Y_k)\\ &= (\pi^*\psi)_p(Y_1,\cdots,Y_k).\end{aligned}$$

□